MAPUA MALAYAN COLLEGES MINDANAO

SmartNutri Scanner

An AI-Powered Nutrition Monitoring and Meal Recommendation System for Early Childhood Malnutrition Prevention.

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ABSTRACT

SmartNutri Scanner is an AI-powered mobile nutrition monitoring and meal recommendation system developed to address the lack of accessible, Filipino-focused dietary tracking tools and the continued risk of childhood malnutrition in the Philippines. Conventional nutrition trackers rely on Western food databases and tedious manual entry, making local dishes such as Sinigang, Adobo, and Kare-Kare difficult to log accurately. The proposed system allows users to capture or describe a meal and instantly receive an estimate of its nutritional content—calories, protein, carbohydrates, fat, fiber, sugar, and sodium—through an artificial-intelligence pipeline. The application was built with React Native and Expo for cross-platform deployment and integrates Google Gemini, Roboflow, Groq, OpenRouter, and OpenAI in an automatic failover chain that sustains reliability when an individual provider becomes unavailable, with a local Filipino-dish nutrition table as the final fallback. Personal meal records are stored on the device by default to protect user privacy, while community features and image hosting use Firebase and Cloudinary. The system also provides daily goal tracking, weekly trends, meal reminders, and child growth monitoring based on World Health Organization (WHO) anthropometric standards. The application was evaluated through functional testing and a useracceptance survey based on the ISO/IEC 25010 software-quality model. Results showed that all defined functional requirements were implemented and verified, that the application reliably recognized common Filipino dishes with reasonable nutrient estimates, and that respondents rated the system positively for functional suitability, usability, and reliability. SmartNutri Scanner demonstrates that an affordable, locally relevant, AI-assisted nutrition tool can support better dietary awareness among Filipino families.

Keywords — nutrition monitoring, food image recognition, Filipino cuisine, mobile application, artificial intelligence, multi-provider failover, ISO/IEC 25010

CHAPTER I: INTRODUCTION

1.1 Background of the Study

Proper nutrition plays a critical role in maintaining health and preventing dietrelated illnesses. According to the World Health Organization (WHO), unhealthy diets are among the leading risk factors for noncommunicable diseases such as diabetes, cardiovascular diseases, and obesity (WHO, 2023). In the Philippines, where traditional cuisine forms a core part of daily life, many Filipinos struggle to monitor the nutritional content of their meals. Most food tracking applications available today are designed around Western food databases and do not include accurate data for common Filipino dishes such as Sinigang, Adobo, Kare-Kare, and Tinola. Because of this, many Filipino users find it difficult to track their daily nutrition accurately using existing tools.

At the same time, mobile technology and artificial intelligence are becoming increasingly accessible. Smartphones are widely used across the country, with the Philippines maintaining one of the highest rates of internet and mobile usage in Southeast Asia (DataReportal, 2024). Furthermore, advances in artificial intelligence, particularly in computer vision and image recognition, have enabled systems to identify food items and estimate nutritional information from images with increasing accuracy (Jelodar et al., 2023). These technologies present an opportunity to build a more relevant and practical nutrition tracking tool designed specifically for Filipino food culture.

On a global scale, research shows that AI-based food recognition systems can identify dishes from photographs and estimate their nutritional content with reasonable accuracy. Mezgec and Koroušić Seljak (2017) explained that deep learning models trained on food image datasets can classify food items and provide nutrient estimates as part of dietary monitoring tools. These findings confirm that image-based food analysis is a viable approach for building nutrition tracking systems.

In the local context, studies in the Philippines highlight the nutritional challenges faced by communities. Disa et al. (2022) found that many Filipinos have limited knowledge of the nutritional content of the foods they eat every day. Another study by Dela Cruz and Reyes (2021) found that children in Davao City are at risk of micronutrient deficiencies, particularly in iron, zinc, and vitamin A. These findings show that there is a clear need for tools that can help Filipinos understand and monitor what they consume.

Because of this, the researcher developed SmartNutri Scanner, an AI-powered mobile application that allows users to take or upload a photo of a meal and instantly receive an analysis of its nutritional content. The system identifies the food items, estimates macronutrient values, and allows the user to log the meal to their daily history. The application stores all personal meal data locally on-device as the primary source of truth, and employs a multi-provider AI architecture with automatic failover across Google Gemini, Roboflow, Groq, OpenRouter, and OpenAI to ensure reliable nutritional analysis. Authentication is handled through Firebase, and optional cloud synchronization supports community features and optional meal backup.

Although previous studies have discussed AI-based food recognition and general nutrition tracking, there is limited research focused on a locally adapted nutrition tracker that prioritizes on-device data storage, integrates a multi-provider AI failover architecture, and incorporates child malnutrition monitoring designed specifically for Filipino food culture. SmartNutri Scanner aims to fill this gap.

1.2 Statement of the Problem

Nutrition tracking remains a significant challenge for many Filipinos, particularly those who rely on traditional home-cooked meals and carinderia food that are not represented in existing food databases. Most available nutrition applications such as MyFitnessPal, Lose It!, and Cronometer are designed around Western food databases and do not support accurate analysis of common Filipino dishes such as Sinigang, Adobo, Kare-Kare, and Tinola. As a result, health-conscious Filipinos, underweight individuals, and parents monitoring their children's dietary intake cannot accurately track their daily nutritional consumption using existing tools.

Beyond personal dietary tracking, child malnutrition remains a critical public health concern in the Philippines. Dela Cruz and Reyes (2021) found that children in Davao City are at risk of iron, zinc, and vitamin A deficiencies linked to poor dietary monitoring and limited access to nutrition education tools. Parents and caregivers have no accessible, locally relevant, and AI-powered tool to help them track whether their children are meeting daily nutritional requirements.

Therefore, this study aimed to develop SmartNutri Scanner, an AI-powered mobile nutrition monitoring and meal recommendation system. Specifically, the study sought to answer the following questions:

- How can the Google Gemini multimodal AI model identify Filipino food items from photographs and return estimated nutritional data including calories, protein, carbohydrates, fat, fiber, sugar, and sodium?
- How can a multi-provider AI failover architecture be implemented to ensure reliable AI-powered food analysis across Google Gemini, Roboflow, Groq, OpenRouter, and OpenAI, with automatic fallback and a local Filipino-dish nutrition table as the final fallback?
- How can a meal logging and history system be developed to record daily food intake, group meals by date, and display weekly nutrition trends through chart visualizations?
- How can child nutrition monitoring be integrated to allow parents and caregivers to track the dietary intake of their children and identify micronutrient deficiencies?
- How can supporting features including daily nutrition goal tracking, meal reminders, BMR-based goal calculation, and data export and import improve the overall usability and practicality of the application?
- How effective is the SmartNutri Scanner application based on ISO software quality standards in terms of: (a) Functionality Suitability; (b) Performance Efficiency; (c) Usability; (d) Reliability; (e) Security; and (f) Portability?

1.3 Objectives of the Study

The study aimed to develop a mobile nutrition tracking application that uses AI to analyze meals from photographs and help users monitor their daily dietary intake. Specifically, the study aimed to:

- Implement an AI-powered food scanner that identifies meals from photographs using the Google Gemini API and returns estimated nutritional data including calories, protein, carbohydrates, fat, fiber, sugar, and sodium.
- Develop a meal logging and history system that records daily food intake, groups meals by date, and displays weekly nutrition trends through chart visualizations.
- Build supporting features including daily nutrition goal tracking with a calorie progress bar and macronutrient summary pills, meal reminders via local notifications, a profile

setup wizard with BMR-based goal calculation, and data export and import functionality.

- Implement a multi-provider AI architecture with automatic failover across Google Gemini (primary), Roboflow (vision fallback), Groq, OpenRouter, and OpenAI, supported by a Firebase Cloud Functions backend proxy and a local Filipino-dish nutrition table as the final fallback, to ensure reliable AI-powered food analysis and system resilience.
- Integrate child nutrition monitoring features that allow parents and caregivers to create child profiles, track daily dietary intake, and monitor growth status using WHO anthropometric growth standards (weight-for-age, height-for-age, and weight-for-height z-scores) to flag malnutrition risk.
- Evaluate the acceptability of the proposed SmartNutri Scanner application based on ISO software quality standards in terms of Functionality Suitability, Performance Efficiency, Usability, Reliability, Security, and Portability.

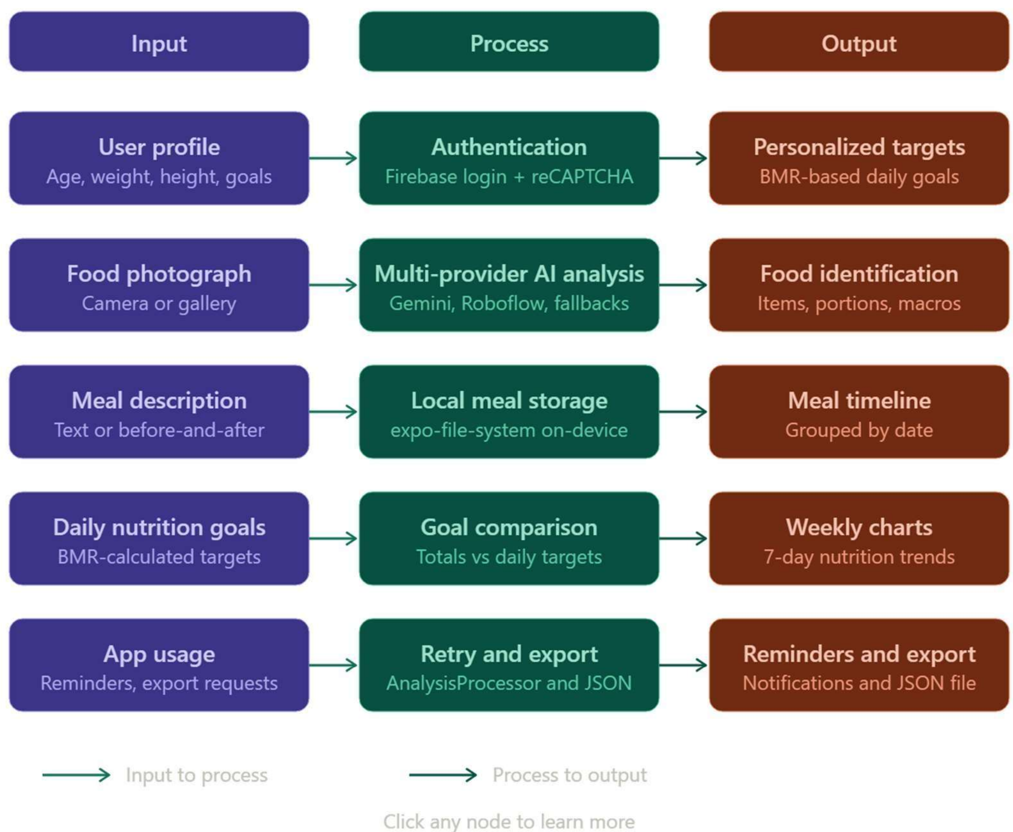
1.4 Conceptual Framework

SmartNutri Scanner follows an Input-Process-Output (IPO) conceptual framework. The IPO framework served as the foundation for organizing the system development activities, AI processing procedures, data storage workflows, and evaluation processes integrated throughout the implementation of the proposed mobile application.

The Input component consists of user profile data including age, gender, weight, height, and goals; food photographs captured from the device camera or selected from the gallery; optional meal comments or descriptions; before and after meal photos; and daily nutrition goals. These inputs represent the primary data sources the system collects from the user to perform nutritional analysis and monitoring.

The Process component involves the core operations of the system including user authentication via Firebase, sending food images to the multi-provider AI service (Google Gemini primary, with Roboflow, Groq, OpenRouter, OpenAI, and a local Filipino-dish nutrition table as fallbacks), parsing the JSON nutritional response, storing meal records to the local file system, calculating daily nutrition totals, comparing totals against user goals, retrying failed analyses via the AnalysisProcessor background service, and exporting or importing data as JSON files.

The Output component consists of the developed SmartNutri Scanner mobile application delivering identified food items with serving sizes; macronutrient breakdowns covering calories, protein, carbohydrates, fat, fiber, sugar, and sodium; health benefits and concerns; a daily calorie progress bar with macronutrient summary pills; a meal timeline grouped by date; weekly nutrition history charts; exportable JSON data files; and meal reminders delivered via local notifications. Table 1. Input-Process-Output Framework of SmartNutri Scanner



1.5 Scope and Limitations of the Study

Scope

SmartNutri Scanner covers AI-powered food identification and nutritional analysis for any meal photographed by the user, with no restriction to a fixed food database. The application supports single-image analysis, before-and-after comparison analysis, text-only meal logging, meal correction via the Fix It feature, and meal relogs. The system is developed as a cross-platform mobile application using React Native with Expo, targeting Android and iOS devices.

All personal meal data, user profiles, and daily goals are stored locally ondevice using expo-file-system as the primary source of truth. A Firebase Cloud Functions backend provides server-side AI proxy endpoints (aiProxy, roboflowProxy, verifyTurnstile), and Firebase Firestore stores community posts, user profiles, and food-recognition learning data. Authentication is handled through Firebase using email/password or Google sign-in. Supporting features include daily nutrition goal tracking, BMR-based goal calculation, weekly nutrition charts, meal reminders, Health Connect sync on Android, and JSON data export and import.

Limitations

Nutritional values returned by the AI providers are estimates based on general food knowledge and may not precisely match laboratory-measured values for specific portion sizes or regional recipe variations. The AI scanning feature requires an active internet connection, and offline analysis is not supported in the current version. Ratelimiting risks are mitigated by the multi-provider failover architecture, which automatically advances through multiple API keys and providers (Google Gemini, Roboflow, Groq, OpenRouter, OpenAI) before falling back to a local Filipino-dish nutrition table; however, sustained heavy usage may still encounter temporary throttling across all providers simultaneously.

Image quality and lighting conditions affect the accuracy of food identification, and blurry or poorly lit photographs may result in incomplete or incorrect analysis. The application does not include barcode scanning for packaged food products. Health Connect sync is available on Android only. The system was developed and tested by a solo developer and has not undergone large-scale pilot deployment or formal clinical validation.

1.6 Significance of the Study

The findings of this study are expected to contribute to the advancement of nutrition monitoring systems through the integration of artificial intelligence, mobile computing, and locally relevant food analysis tools. The proposed system may provide practical and technological benefits to the following stakeholders:

Health-Conscious Filipinos and College Students

The proposed system may assist health-conscious Filipinos, underweight individuals, and college students in tracking their daily nutritional intake through AIpowered meal scanning. By eliminating the need for manual food entry and providing instant nutritional analysis from photographs, SmartNutri Scanner reduces the effort required for daily nutrition monitoring and encourages healthier dietary habits among young Filipinos.

Parents and Caregivers

The proposed system may assist parents and caregivers in monitoring the dietary intake of their children and screening for growth-related malnutrition risk associated with micronutrient deficiencies such as iron, zinc, and vitamin A. Through child nutrition profiles, WHO-based growth tracking, and daily nutritional summaries, the application provides an accessible tool for supporting early childhood malnutrition prevention within Filipino households.

Barangay Health Workers and Nutritionists

The proposed system may support barangay health workers, nutritionists, and public health practitioners by providing a technology-driven tool capable of assisting dietary monitoring and nutrition education activities within local communities. The application may serve as a supplementary reference tool during nutrition counseling and health monitoring activities.

Information Technology and Emerging Technology Practitioners

The study may serve as a reference for information technology professionals and software developers interested in implementing AI-powered mobile applications using the Google Gemini multimodal model, a multi-provider AI failover architecture, React Native, Expo, and Firebase technologies within real-world health monitoring systems.

Researchers and Future Developers

The study may provide future researchers and developers with baseline information, implementation methodologies, and development references related to AI-powered nutrition tracking, Filipino food analysis, child malnutrition monitoring, and privacy-first mobile application development. The findings may serve as a foundation for future studies involving larger datasets, multi-language support, IoT integration, and clinical validation procedures.

Students and Academic Institutions

The study may contribute to academic institutions and students within the fields of Computer Science, Information Technology, Applications Development, and Emerging Technologies by demonstrating the practical integration of artificial intelligence and mobile computing within health monitoring applications.

1.7 Definition of Terms

The following terms are operationally and conceptually defined to provide a clearer understanding of the concepts, technologies, and terminologies utilized in the study:

Artificial Intelligence (AI)

Artificial Intelligence refers to the simulation of human intelligence processes through computer systems capable of performing tasks such as learning, reasoning, decision-making, and pattern recognition. In this study, artificial intelligence refers to the integration of the Google Gemini multimodal large language model utilized for identifying food items from photographs and estimating their nutritional composition.

Basal Metabolic Rate (BMR)

Basal Metabolic Rate refers to the number of calories the human body requires to maintain basic physiological functions at rest. In this study, BMR refers to the calculation used by the profile setup wizard to determine the user's recommended daily caloric and macronutrient intake targets based on age, gender, height, weight, and activity level.

Calorie Tracking

Calorie tracking refers to the process of recording and monitoring the caloric content of foods consumed by a user on a daily basis. In this study, calorie tracking refers to the daily nutrition monitoring feature of SmartNutri Scanner that aggregates macronutrient totals from logged meals and compares them against the user's daily nutrition goals.

Expo

Expo is an open-source framework and platform built on top of React Native that simplifies the development, building, and deployment of cross-platform mobile applications for Android and iOS. In this study, Expo refers to the primary development framework utilized in building the SmartNutri Scanner mobile application.

Firebase

Firebase is a cloud-based platform developed by Google that provides database management, cloud storage, authentication, and real-time synchronization functionalities. In this study, Firebase refers to the cloud platform utilized for storing community posts, user profiles, and shared data associated with the community features of the application.

Google Gemini API

Google Gemini API refers to the multimodal large language model API developed by Google that is capable of processing both image and text inputs to generate structured responses. In this study, the Google Gemini API refers to the primary AI service utilized by SmartNutri Scanner to identify food items from photographs and return estimated nutritional data in structured JSON format.

Macronutrients

Macronutrients refer to the primary nutritional components required by the human body in large quantities, including calories, protein, carbohydrates, fat, fiber, sugar, and sodium. In this study, macronutrients refer to the nutritional values identified and tracked by the SmartNutri Scanner application for each logged meal.

Micronutrients

Micronutrients refer to vitamins and minerals required by the human body in small quantities for proper growth, development, and physiological function. In this study, micronutrients refer to essential nutrients including iron, zinc, and vitamin A whose deficiencies are associated with childhood malnutrition in the Philippines.

React Native

React Native is an open-source mobile application development framework developed by Meta that enables developers to build cross-platform applications for Android and iOS using JavaScript and React. In this study, React Native refers to the primary programming framework utilized in building the SmartNutri Scanner mobile application.

expo-secure-store expo-secure-store is a hardware-backed secure storage library within the Expo ecosystem that encrypts and stores sensitive data on the device. In this study, exposecure-store refers to the secure storage mechanism utilized to protect the user's authentication token on-device.

expo-file-system expo-file-system is a library within the Expo ecosystem that provides access to the device's local file storage for reading, writing, and managing files. In this study, expo-file-system refers to the local storage solution utilized by SmartNutri Scanner to save meal records, images, and user data entirely on-device without requiring a backend server.

CHAPTER II: REVIEW OF RELATED LITERATURE AND TECHNOLOGIES

This chapter presents the related literature, technologies, and studies relevant to the development of the proposed AI-powered nutrition monitoring and meal recommendation system. The discussions focus on food recognition using artificial intelligence, Filipino dietary challenges, child malnutrition, mobile health applications, and on-device data storage design. The chapter establishes the technological and conceptual foundations supporting the development of SmartNutri Scanner and identifies the research gaps addressed by the study.

2.1 AI-Based Food Recognition Systems

Advances in large language models have made it possible to identify food items from photographs and describe their nutritional composition. Mezgec and Koroušić Seljak (2017), in the study NutriNet: A Deep Learning Food and Drink Image Recognition System for Dietary Assessment, demonstrated that models trained on food image datasets can classify food items and provide nutritional estimates useful for dietary monitoring. The study showed that automated food recognition reduces the manual effort required for tracking meals and increases the consistency of nutritional records.

This finding supports SmartNutri Scanner because the application uses the Google Gemini multimodal model to identify food items from user-uploaded photographs and return estimated nutritional values in structured JSON format. Instead of requiring users to manually search and select food items from a database, the system automates the identification process, which reduces errors and saves time. A more recent benchmark study by NutriBench (2024) further demonstrated that large language models can now estimate nutritional content from meal descriptions with increasing accuracy, supporting the use of general-purpose AI models for food analysis tasks.

2.2 Filipino Nutrition and Dietary Monitoring

Research in the Philippines consistently highlights nutritional awareness as a challenge for many households. Disa et al. (2022), in their study on nutrition knowledge among Filipino adults, found that many respondents are unaware of the nutritional composition of common foods they consume daily. The lack of accessible tools contributes to this gap because most available nutrition apps do not cover local dishes.

This finding directly supports the need for SmartNutri Scanner. Because Filipino dishes are not well represented in international food databases, users who rely on general apps often receive inaccurate nutritional data. SmartNutri Scanner uses a general-purpose AI model rather than a fixed database, allowing it to analyze any food a user photographs without being limited to a predefined list.

2.3 Micronutrient Deficiency Among Filipino Children

Micronutrient deficiency remains a significant health concern for Filipino children. Dela Cruz and Reyes (2021) found that children in Davao City face risks of iron, zinc, and vitamin A deficiency, which are linked to poor concentration, weakened immunity, and delayed physical development. These deficiencies are preventable through proper dietary monitoring and education.

This finding supports the design of SmartNutri Scanner as a locally relevant nutrition tool. The application is built specifically for Filipino users and acknowledges the micronutrient concerns prevalent in the local context. While the current version focuses on macronutrient tracking, the architecture is designed to support expansion into micronutrient monitoring in future iterations.

2.4 Mobile Applications for Health and Nutrition Tracking

The use of mobile applications for health tracking has grown significantly in recent years. Lieffers and Hanning (2012), in Dietary Assessment and Self-Monitoring with Nutrition Applications for Smartphones, reviewed several nutrition tracking apps and found that mobile tools increase users' awareness of their dietary habits and encourage healthier food choices. The review also noted that ease of use and visual feedback are critical factors in whether users continue using a nutrition application long-term.

This finding supports the design decisions made for SmartNutri Scanner. The application uses a camera-first interface to simplify meal logging and provides visual progress indicators including a calorie progress bar with macronutrient summary pills and weekly trend charts. These features are aligned with the characteristics identified in the review as factors that improve long-term user retention.

2.5 Privacy and On-Device Data Storage

Growing concerns about user privacy in health applications have led researchers to explore local and on-device storage as an alternative to cloud-based architectures. Eysenbach (2008), in Medicine 2.0: Social Networking, Collaboration, Participation, Apomediation, and Openness, discussed the role of trust, user control, and open access in health-related digital tools, noting that user engagement is closely tied to perceived control over personal data.

SmartNutri Scanner addresses data stewardship directly by storing all personal meal history, user profiles, and daily goals in the device’s local file system using expofile-system as the primary source of truth. Personal meal images remain on-device and are never uploaded to cloud storage; community post images are uploaded to Cloudinary, a third-party image host, solely to enable the optional community sharing feature. A Firebase Cloud Functions backend handles AI proxying server-side so that API keys are not embedded in client-side code. Firebase

Authentication secures community access, and Firestore stores only community posts, user profiles, and food-recognition learning data. The authentication token is protected in expo-securestore. This architecture balances user data sovereignty for personal nutrition records with the cloud infrastructure required for community and AI proxy features.

2.6 Synthesis of Reviewed Literature and Technologies

The reviewed literature and technologies established that nutrition tracking remains a significant challenge for Filipino users because most existing applications are built around Western food databases that do not support accurate analysis of local dishes. Studies by Disa et al. (2022) and Dela Cruz and Reyes (2021) confirmed that nutritional awareness gaps and micronutrient deficiencies continue to affect Filipino households, particularly among children in Davao City. Consequently, there is a clear need for a locally relevant, accessible, and AI-powered nutrition monitoring tool designed specifically for Filipino food culture.

The reviewed studies also demonstrated that AI-based food recognition systems using deep learning and multimodal large language models are highly effective in identifying food items from photographs and estimating their nutritional composition. Mezgec and Koroušić Seljak (2017) confirmed that automated food recognition reduces manual effort and improves consistency in dietary monitoring. Furthermore, Lieffers and Hanning (2012) identified ease of use and visual feedback as critical factors in the long-term adoption of nutrition tracking applications.

Despite these advancements, several research gaps remain evident in existing nutrition monitoring systems. Most applications require manual food entry, rely on fixed databases that do not include Filipino dishes, and require paid subscriptions that limit accessibility. Existing systems also do not address child malnutrition monitoring within a locally adapted mobile application framework that prioritizes on-device data storage. Furthermore, limited studies focus on the integration of multi-provider AI failover architectures within nutrition tracking systems that maintain personal data ondevice while leveraging cloud infrastructure for community features and AI proxying.

Therefore, the present study addresses these gaps through the development of SmartNutri Scanner, an AI-powered mobile application integrating a multi-provider Google Gemini-led AI architecture, Firebase Cloud Functions backend proxy, React Native with Expo, local file storage as the primary data source of truth, Firebase community features, and child nutrition monitoring functionalities designed specifically for Filipino users and early childhood malnutrition prevention.

2.7 Matrix Comparison of Related Literature and Studies

Table 2. Matrix Comparison of Related Literature and Studies

Author / Year	Title / Study Focus	Technology Used	Key Findings	Identified Gaps	Relation to Present Study
Mezgec & Koroušić Seljak (2017)	NutriNet: Deep Learning Food Recognition	Deep Neural Networks, Image Classification	Automated food recognition reduces manual effort and improves dietary monitoring consistency	Limited to controlled food datasets; no Filipino food support	Supports use of AI for food identification from photographs
Lieffers & Hanning (2012)	Dietary Assessment with Nutrition Apps	Mobile app review	Ease of use and visual feedback improve long-term app adoption	Reviewbased; no AI integration	Supports camera-first interface and visual progress indicators
Disa et al. (2022)	Nutritional Knowledge Among Filipino Adults	Surveybased study	Many Filipinos unaware of nutritional content of daily foods	No technologybased solution proposed	Supports need for locally relevant nutrition tracking tool
Dela Cruz & Reyes (2021)	Micronutrient Deficiency in Davao City Children	Crosssectional study	Children in Davao at risk of iron, zinc, vitamin A deficiency	No mobile monitoring tool proposed	Supports child nutrition monitoring feature of SmartNutri Scanner

Author / Year	Title / Study Focus	Technology Used	Key Findings	Identified Gaps	Relation to Present Study
Eysenbach (2008)	Medicine 2.0 and Health Applications	Web and mobile health systems	Mobile and social technologies improve health awareness and user engagement	Early study; no AI or LLM integration	Supports mobile-first approach to health and nutrition monitoring
NutriBench (2024)	LLM Nutrition Estimation Benchmark	Large Language Models, NLP	LLMs can estimate nutrition from meal descriptions with measurable accuracy	Not focused on photobased analysis for Filipino food	Supports use of LLMs such as Gemini for nutritional analysis
Present Study	SmartNutri Scanner	Google Gemini API, React Native, Expo, Firebase, multiprovider AI failover	AI-powered Filipino meal scanning, child nutrition monitoring, privacy-first local storage	Addresses Filipino food gap, child malnutrition, privacy-first on-device storage	Addresses all identified gaps through AI integration and local food relevance

2.8 Matrix Comparison of Related Systems

Table 3. Matrix Comparison of Related Systems

Feature	MyFitnessPal	Lose It!	Cronometer	SmartNutri Scanner
AI Photo Scanning	No	No	No	Yes (Gemini API)
Filipino Food Support	No	No	No	Yes
Account Required	No	No	No	Yes
Free / No Subscription	No	No	No	Yes (free; developermanaged API keys)

Child Nutrition Monitoring	No	No	No	Yes
On-Device Data Storage	No	No	No	Yes
Before and After Meal Analysis	No	No	No	Yes
Community Posts	No	No	No	Yes
Recipe Index	No	Limited	No	Yes
Data Export / Import	Limited	Limited	Yes	Yes
Meal Cost Estimation	No	No	No	Yes

CHAPTER III: PROJECT METHODOLOGY AND SYSTEM PLANNING

3.1 Research Design

The study utilized a developmental and implementation-oriented research design focusing on the integration of artificial intelligence, mobile computing, and emerging technologies for nutrition monitoring and early childhood malnutrition prevention. The developmental research approach is commonly utilized in software engineering and Applications Development and Emerging Technologies because it emphasizes system planning, technology integration, iterative development, deployment implementation, and evaluation procedures associated with softwarebased solutions (Richey and Klein, 2007).

The study specifically focused on the development of SmartNutri Scanner, an AI-powered mobile application integrating a multi-provider Google Gemini-led AI architecture with automatic failover, Firebase Cloud Functions backend proxy, React Native with Expo, local file storage, Firebase cloud services, and child nutrition monitoring functionalities. The research design emphasized AI-powered food identification, meal logging, nutritional analysis, community features, and mobile deployment utilizing locally relevant food analysis capabilities.

The proposed system was evaluated using ISO software quality standards to assess the acceptability of the application in terms of functionality suitability, performance efficiency, usability, reliability, security, and portability.

3.2 System Development Methodology

SmartNutri Scanner was developed using the Agile software development methodology, structured into iterative sprints aligned with the course module requirements. The Agile approach was chosen because it allows continuous testing and adjustment throughout the development process rather than requiring all features to be finalized before any testing begins. Each sprint produced a working increment of the system that could be evaluated and refined before the next sprint began.

1.0 Project Kick-Off

The first phase established the overall direction of the project. The researcher identified the core problem, that health-conscious Filipinos and underweight individuals have no accessible AI-powered tool to track nutrition from local food photos, and proposed SmartNutri

Scanner as the solution. The project title, target users, rationale, and initial concept were defined. A multi-provider AI architecture with on-device local storage as the primary data source was chosen at this stage to ensure reliable food analysis and data accessibility even under adverse network conditions, while Firebase cloud services were designated for authentication, community features, and optional meal backup.

2.0 Requirements Gathering

After the project direction was established, all necessary information for building the system was collected and documented. Functional and non-functional requirements were identified, and the technology stack was finalized. React Native with Expo was selected for cross-platform mobile development. Google Gemini was chosen as the primary AI provider due to its multimodal capabilities, with Roboflow, Groq, OpenRouter, and OpenAI designated as fallback providers in an automatic failover chain. Firebase was selected for authentication, cloud proxy functions, and community data. expo-file-system was selected for local meal storage, and exposecure-store for secure authentication token management. Initial wireframes were created to visualize the Home, Scan/Add, Services/Growth, Community, and Profile screens.

3.0 System Design

The system architecture was designed around three core layers: the UI layer built with React Native components, the service layer handling business logic, and the storage layer managing local file persistence. The tab-based navigation structure was defined with five primary tabs: Home (daily nutrition summary and meal timeline), Services/Growth (WHO growth charts and services hub), Scan/Add (center-button AI food scanning), Community (posts and comments), and Profile (settings, goals, and account). A modal-first design pattern was chosen for adding meals and viewing meal details, keeping the main navigation clean. The Gemini prompt structure was designed with four distinct system prompts covering single-image analysis, before-and-after comparison, text-only analysis, and meal correction.

4.0 Core Development

Development was carried out incrementally across multiple weeks. The onboarding flow was built first to handle user profile setup and Firebase authentication (email/password and Google sign-in with reCAPTCHA). The multi-provider AI service was then implemented, with a GeminiService handling all four analysis modes (singleimage, before-and-after, text-only, Fix It) using a failover chain through Gemini models (gemini-2.5-flash primary, gemini-2.0-flash, gemini-2.5-flash-lite, gemini-2.0-flash-lite, gemini-2.5-pro), Roboflow vision classification, Groq, OpenRouter, OpenAI, and a local Filipino-dish nutrition table as the final fallback. The Firebase Cloud Functions backend was implemented with three HTTPS endpoints: aiProxy, roboflowProxy, and verifyTurnstile, enabling both proxy mode and direct mode operation. The FileSystemStorageService was built to handle all local data operations including saving, updating, deleting, and retrieving meal records. The AddMealModal was implemented with camera capture, gallery selection, before-and-after mode, and a text-only skip option. The AnalysisProcessor was built as a background service that handles pending analyses, retries failed requests, and resumes interrupted analyses when the app returns to the foreground. The MealDetailModal was implemented with editable nutrition fields, title editing, date and time editing, a Fix It AI correction feature, and a relog function.

5.0 Integration and Testing

All modules were integrated and tested systematically. The Health Connect integration was implemented for Android to sync nutrition data with the system health platform. Meal reminders were integrated using expo-notifications with daily scheduled alerts. Unit tests were written for the ExportImportService covering the Storage Access Framework flow on Android, iOS sharing fallback, import schema validation, and base64 image conversion. Functional testing was conducted for all user-facing features. Bug fixes addressed scroll freezing on Android modal components, duplicate meal entries in the event system, notification rescheduling after the app wakes from background, and SAF permission denial fallback during export.

6.0 Deployment and Documentation

The final application was packaged using EAS Build targeting Android SDK 35. Technical documentation was prepared including a README with setup instructions, architecture overview, and service descriptions. The open-source license was embedded in the application profile screen along with the font license. The application was prepared for demonstration covering the complete user journey from onboarding through meal scanning, nutrition review, goal tracking, and data export.

3.3 System Requirements Engineering

3.3.1 Functional Requirements

The following functional requirements define the specific behaviors and features that SmartNutri Scanner must support:

Table 4. Functional Requirements of the Proposed System

ID	Requirement	Description
FR01	User Onboarding	The system shall guide new users through a multi-step profile setup wizard collecting age, gender, height, weight, activity level, and goal to calculate BMR-based daily nutrition targets.
FR02	Authentication and Secure Token Storage	The system shall authenticate users via Firebase using email/password or Google sign-in, protected by Google reCAPTCHA. The user's authentication token shall be stored securely on-device using exposecure-store. Developer API keys for all AI providers shall be managed server-side via Firebase Cloud Functions and via environment configuration, and shall not be entered by the user.
FR03	AI Provider Failover	The system shall route food-analysis requests through a fixed, developermanaged model and provider failover chain (Google Gemini models, then Roboflow, Groq, OpenRouter, and OpenAI), automatically advancing to the next option upon failure without requiring the user to configure or select a model.
FR04	AI Food Scanning (Photo)	The system shall allow users to take a photo using the device camera or select one from the gallery, then send it to the Gemini API and display identified food items with nutritional breakdown.
FR05	AI Food Scanning (Text)	The system shall allow users to skip photo capture and instead describe their meal in text, which is sent to Gemini for analysis using a text-only system prompt.
FR06	Before and After Analysis	The system shall allow users to capture a before photo and an after photo of the same meal, then send both to Gemini to calculate nutrition only for the consumed portion.
FR07	Meal Logging	The system shall save analyzed meals to local file storage with a timestamp, image URI, nutritional analysis, and optional comment.
FR08	Meal Timeline	The system shall display all logged meals grouped by date on the home screen, sorted with the most recent date first.
FR09	Meal Detail and Editing	The system shall allow users to view full nutritional details of any logged meal and directly edit nutrient values, the meal title, and the timestamp.

ID	Requirement	Description
FR10	Fix It AI Correction	The system shall allow users to submit a correction comment on a logged meal, which is re-sent to Gemini with the original image and the current analysis to generate a revised nutritional result.
FR11	Daily Nutrition Tracker	The system shall display daily totals for calories, protein, fat, carbohydrates, fiber, sugar, and sodium with a calorie progress bar and macronutrient summary pills comparing current intake to the user's daily goals.
FR12	Nutrition History Charts	The system shall display weekly line charts for each macronutrient on the Profile screen, showing the last 7 days of daily totals against the user's goal.
FR13	Daily Goals Editor	The system shall allow users to manually edit their daily nutrition targets for calories, protein, fats, carbs, fiber, sugar, and sodium, or recalculate them using the onboarding profile wizard.
FR14	Meal Reminders	The system shall allow users to create, enable, disable, and delete meal reminder notifications with custom names and times using exponotifications.
ID	Requirement	Description
FR15	Data Export	The system shall export all meal history, user profile, daily goals, meal reminders, and AI model preference as a single JSON file with base64encoded images, using Android SAF or iOS sharing.
FR16	Data Import	The system shall import a previously exported JSON file, validate its schema, restore the user profile and settings, and copy meal images back to local storage.
FR17	Meal History Management	The system shall allow users to delete individual meals or clear meal history by time range (past hour, day, month, year, or all).
FR18	Analysis Retry	The system shall automatically retry failed meal analyses when the app returns to the foreground or network connectivity is restored, and allow manual retry from the UI.
FR19	Health Connect Sync	The system shall write nutrition records to Android Health Connect when permission is granted, using the meal timestamp and macronutrient totals.
FR20	Relog Meal	The system shall allow users to relog a previously analyzed meal with the current timestamp, creating a new meal entry with the same analysis and images.
FR-21	AI Food Assistant Chatbot	The system shall provide a conversational AI assistant, accessible from the Services hub, that answers nutrition and Filipino-food questions using a Groq-primary multi-provider failover chain.

3.3.2 Non-Functional Requirements

Table 5. Non-Functional Requirements of the Proposed System

Category	Requirement
Usability	The application shall be operable by non-technical users. Core functions (scan, view, delete) shall be accessible within two taps from the home screen. The onboarding flow shall require no prior knowledge of nutrition science.
Performance	The AI food scanning feature shall display a loading indicator immediately after the user saves a meal, and the analysis result shall appear as soon as the Gemini API responds. Meal history shall load within one second on devices with up to 50 stored entries.
Reliability	The system shall not lose meal data if the app is closed during or after analysis. Pending analyses shall be resumed automatically on next launch via AnalysisProcessor. The local storage index shall remain consistent with stored meal files.
Security	Developer API keys for all AI providers shall be managed via a Firebase Cloud Functions proxy where deployed, or otherwise via bundled environment configuration, and shall not be entered by or visible to end users. The user's authentication token shall be stored in expo-secure-store, a hardware-backed secure storage vault. No personal meal data shall be transmitted to any cloud server other than community posts and optional meal backup.
Privacy	All personal user data including meal history, profile, and goals shall be stored ondevice as the primary source of truth. Firebase cloud services shall be used exclusively for authentication, community features, AI proxying, and optional meal backup (off by default). Personal meal images shall remain on-device and shall not be uploaded to cloud storage. Community post images shall be uploaded to Cloudinary solely to support the optional community sharing feature. No third-party analytics tracking shall be integrated.
Compatibility	The application shall run correctly on Android (SDK 28 and above) and iOS devices using the Expo Go development client and as a standalone EAS build.
Maintainability	Source code shall be organized into clearly separated screens, components, services, and hooks. Services shall be stateless where possible and testable independently. Business logic shall not be embedded in UI components.
Offline Behavior	The application shall display all previously logged meal data without an internet connection. The AI scanning feature shall display a clear error message when offline rather than failing silently.

3.4 Research Locale

The study was conducted within the context of Filipino dietary practices and nutritional monitoring needs, with particular focus on Davao City, Philippines. Davao City is recognized as one of the major urban centers in Mindanao and serves as a significant contributor to regional economic and social development. The city's diverse food culture, reliance on traditional

Filipino dishes, and documented challenges with childhood malnutrition and micronutrient deficiency make it a highly relevant locale for the development and evaluation of a locally adapted nutrition monitoring application.

Studies conducted within the local context, including Dela Cruz and Reyes (2021), have highlighted that children in Davao City face risks of iron, zinc, and vitamin A deficiencies linked to poor dietary monitoring practices. The absence of accessible and locally relevant nutrition tracking tools within the region further supports the need for the proposed application. The selected locale strengthened the study's alignment with practical nutrition monitoring applications because the proposed system was developed and evaluated using dietary patterns and food preferences commonly observed within Filipino communities in Davao City.

3.5 Respondents of the Study

The respondents of the study consisted of information technology experts, nutrition-aware individuals, parents and caregivers, and selected end-users who participated in the evaluation and assessment procedures associated with the proposed SmartNutri Scanner mobile application.

Information technology experts participated in evaluating the technical functionality, usability, system performance, reliability, portability, and software quality characteristics of the proposed mobile application based on ISO software quality standards. Their evaluation supported the technical validation and engineering assessment of the proposed system.

Selected end-users including health-conscious individuals, college students, and parents and caregivers monitoring their children's dietary intake additionally participated in the usability and acceptability evaluation procedures of the study. Their feedback contributed to evaluating the accessibility, practicality, and usability of the proposed system within mobile deployment environments.

Table 6. Respondent Classification of the Study

Respondent Group	Description	Primary Role in the Study
Information Technology Experts	IT professionals, faculty members, and technical evaluators	Technical evaluation and software quality assessment

Health-Conscious Individuals	Adults actively monitoring their dietary intake	Usability and practical deployment evaluation
Parents and Caregivers	Selected users monitoring children's nutrition	Child nutrition feature evaluation
College Students	Target end-users from academic environments	General usability and accessibility evaluation

3.6 Research Instruments

The study utilized multiple research instruments and evaluation tools necessary for supporting system validation and software quality evaluation procedures associated with the proposed SmartNutri Scanner mobile application. A structured evaluation questionnaire based on ISO software quality standards was utilized to evaluate the acceptability of the proposed system. The evaluation questionnaire assessed the software quality characteristics of the proposed application in terms of functionality suitability, performance efficiency, usability, reliability, security, and portability. The questionnaire utilized a five-point Likert scale to measure the assessment and perception of the respondents regarding the quality and effectiveness of the proposed mobile application.

Table 7. Research Instruments and Technical Tools

Instrument / Tool	Purpose	Function within the Study
SmartNutri Scanner Mobile Application	Main system platform	AI food scanning, nutrition tracking, child monitoring, community features
Smartphone Camera	Image acquisition	Capturing food photographs for AI analysis
Google Gemini API	AI analysis engine	Food identification and nutritional estimation from photographs
expo-secure-store	Secure token storage	Storing user authentication token securely ondevice
expo-file-system	Local data storage	Storing meal records, images, and user data ondevice
Firebase	Cloud platform	Community posts, shared data, and cloud synchronization
ISO-Based Evaluation Questionnaire	System evaluation instrument	Assessing software quality and system acceptability

3.7 Data Gathering Procedure

The study followed a systematic data gathering procedure to support system development, requirements analysis, and evaluation activities associated with the proposed SmartNutri Scanner application. The procedure involved requirements gathering, system design, iterative development, integration testing, and evaluation processes necessary for the development of the mobile-based nutrition monitoring system.

The initial phase involved identifying the core problem through personal experience, literature review, and analysis of existing nutrition tracking applications. The researcher identified the gap in locally relevant, AI-powered, privacy-first nutrition monitoring tools for Filipino users and proposed SmartNutri Scanner as the solution.

After the requirements phase, the system was developed iteratively following an Agile sprint framework. Each sprint produced a working increment of the application that was tested and refined before the next sprint began. Functional testing was conducted for all user-facing features including AI food scanning, meal logging, nutrition goal tracking, child profile monitoring, community posts, and data export and import.

The final phase involved evaluation by information technology experts and selected end-users using an ISO-based structured questionnaire. Respondents assessed the proposed system in terms of functionality, performance, usability, reliability, security, and portability using a five-point Likert scale. The collected evaluation data was analyzed using weighted mean to determine the overall acceptability of the proposed application.

3.8 Software and Hardware Requirements

Table 8. Software Requirements of the Study

Software / Platform	Purpose	Function within the Study
React Native with Expo	Mobile application development	Cross-platform Android and iOS application development
Google Gemini API	AI analysis engine	Food identification and nutritional estimation
expo-secure-store	Secure storage library	On-device authentication token storage
expo-file-system	Local file storage library	On-device meal record and image storage

Firebase	Cloud platform	Community features and cloud data synchronization
expo-notifications	Notification library	Meal reminder scheduling and delivery
EAS Build	Build platform	Android APK and iOS app build generation
GitHub	Version control	Source code management and version tracking
Visual Studio Code	Code editor	Primary development environment

Table 9. Hardware Requirements of the Study

Hardware Component	Purpose
Laptop / Desktop Computer	Application development and testing
Smartphone Device (Android / iOS)	Application installation, image capture, and testing
Internet Connectivity	Google Gemini API calls and Firebase synchronization

3.9 Statistical Treatment of Data

The study utilized descriptive statistical methods to analyze and interpret the evaluation results gathered during the assessment of the proposed SmartNutri Scanner mobile application. The statistical treatments focused on evaluating the acceptability, usability, functionality, and overall software quality of the proposed system based on ISO software quality standards and respondent evaluation results.

The study utilized the weighted mean to analyze and summarize the evaluation results gathered from information technology experts and selected end-users. The weighted mean enabled the researcher to determine the overall assessment of the respondents regarding the functionality, usability, reliability, and acceptability of the proposed application. The formula for weighted mean is expressed as follows:

$$\bar{x} = \Sigma fx / N$$

Where $\bar{x}$ represents the Weighted Mean, Σfx represents the sum of the product of frequency and rating value, and N represents the total number of responses.

The computed weighted mean values were interpreted using predefined verbal interpretation ranges to determine the level of acceptability of the proposed SmartNutri Scanner application.

Table 10. Likert Scale Interpretation

Scale Range	Verbal Interpretation
4.21 – 5.00	Highly Acceptable
3.41 – 4.20	Acceptable
2.61 – 3.40	Moderately Acceptable
1.81 – 2.60	Less Acceptable
1.00 – 1.80	Not Acceptable

3.10 Work Breakdown Structure (WBS)

The study utilized a Work Breakdown Structure (WBS) to organize, manage, and monitor the engineering activities, technical deliverables, and development tasks associated with the implementation of the proposed SmartNutri Scanner application. A Work Breakdown Structure is a project management tool that decomposes major project phases into smaller and manageable tasks to improve project organization, resource allocation, development monitoring, and implementation control (Project Management Institute, 2021).

Table 11. Work Breakdown Structure of the Study

WBS Phase	Major Activities	Expected Deliverables
1.0 Project Kick-off	Project conceptualization, consultation, scope definition	Approved project concept and objectives
2.0 Project Planning	Sprint planning, resource allocation, timeline preparation	Project plans and schedules
3.0 Requirements Engineering	Functional and non-functional requirements gathering	System requirements documentation
4.0 System Design	Architecture design, workflow design, UI/UX design	System design documents and prototypes
5.0 System Development	AI integration, mobile application development, local storage implementation	Functional SmartNutri Scanner application
6.0 Testing and Quality Assurance	Debugging, validation, usability evaluation	Evaluation and testing reports
7.0 Deployment	Mobile deployment and implementation preparation	Deployable application package

8.0 Documentation and Final Deliverables	Manuscript preparation, technical documentation, final reports	Final manuscript and project deliverables

3.11 Project Timeline and Gantt Chart

The study utilized a project timeline and Gantt Chart to organize, schedule, and monitor the implementation activities associated with the development of the proposed SmartNutri Scanner application. The project timeline served as a planning and monitoring tool that enabled the researcher to track engineering progress, sprint activities, technical deliverables, and implementation schedules throughout the Agile development lifecycle. According to the Project Management Institute (2021), project scheduling tools such as Gantt Charts improve project coordination, resource management, and activity monitoring within software engineering and systems development environments.

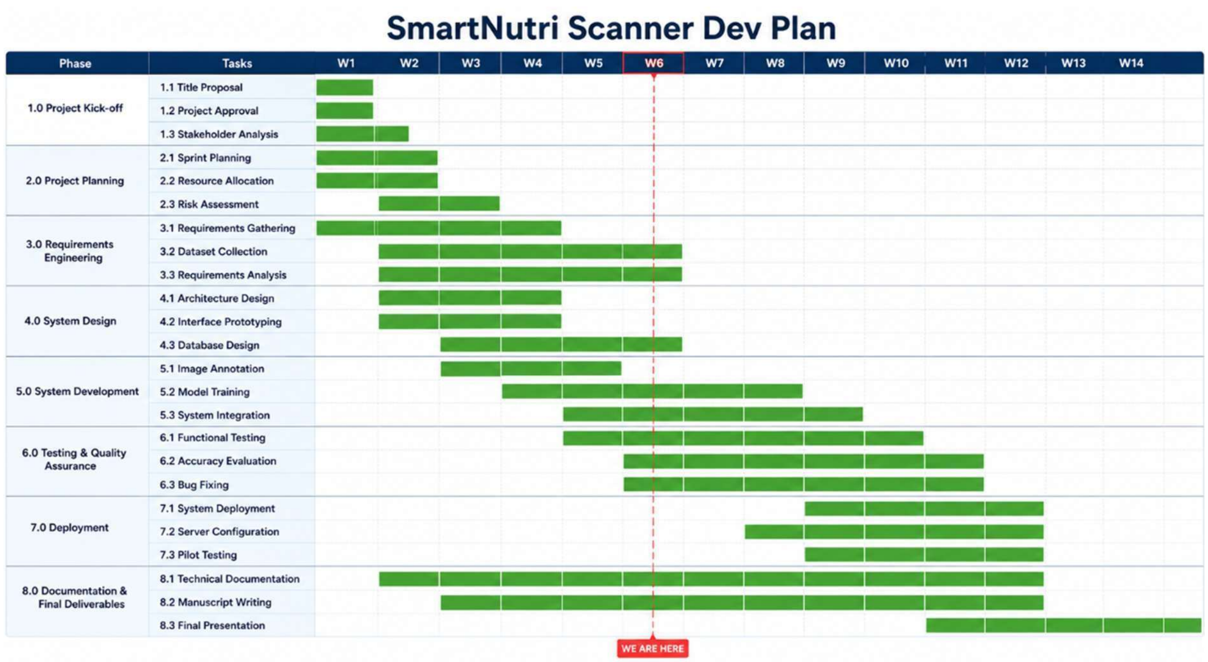


Table 12. Project Timeline of the Study

Agile Phase	Description	Major Activities and Outputs
1.0 Project Kick-off	Identified the research problem, objectives, scope, and proposed technological solutions. Initial concept for SmartNutri Scanner was defined.	Identification of research objectives and scope; technology and tool identification; initial project conceptualization

Agile Phase	Description	Major Activities and Outputs
2.0 Project Planning	Prepared development plans, sprint allocation, scheduling, and project resource management.	Sprint planning; task scheduling; resource allocation; risk identification; project timeline preparation
3.0 Requirements Engineering	Gathered and analyzed system requirements, technology stack, and functional specifications.	Dataset gathering; user requirement analysis; functional and non-functional requirement identification
4.0 System Design	Designed system architecture, workflows, UI/UX, database structures, and Gemini prompt structure.	IPO framework design; mobile interface design; local storage and Firebase design; object detection workflow design
5.0 System Development	Implemented the application incrementally with all modules including onboarding, scanner, tracker, community, and export.	Onboarding flow; GeminiService; FileSystemStorageService; meal logging; AnalysisProcessor; Fix It feature
6.0 Testing and Quality Assurance	Evaluated performance, reliability, and usability using testing procedures and ISObased software quality standards.	Unit testing; functional testing; usability and compatibility testing; bug fixing; User Acceptance Testing
Agile Phase	Description	Major Activities and Outputs
7.0 Deployment	Packaged and deployed the application using EAS Build targeting Android SDK 35.	Android APK build; Firebase production setup; pilot testing; deployment sign-off
8.0 Documentation and Final Deliverables	Prepared technical documentation, research manuscripts, testing reports, and presentation materials.	Technical documentation; research manuscript; user manual; testing result consolidation; final presentation

3.12 Roles and Responsibilities

The development of the proposed SmartNutri Scanner application required the coordination of multiple technical, managerial, and evaluation roles to support project planning, AI integration, mobile application development, cloud synchronization, testing, deployment, and evaluation procedures.

Table 13. Project Roles and Responsibilities

Role ID	Role	Primary Responsibilities
PS	Project Sponsor / Adviser	Provides project supervision, strategic guidance, approval of project milestones, and overall evaluation of project implementation and deliverables

PM	Project Manager	Oversees project planning, sprint management, scheduling, resource allocation, and overall project coordination
SA	Systems Analyst	Analyzes system requirements, prepares workflow analysis, develops functional specifications, and aligns system processes with project objectives
PRG	Programmer / Developer	Develops application functionalities, implements frontend and backend components, integrates Firebase and Gemini API services, and supports debugging
UI	UI/UX Designer	Designs user interfaces, user experience workflows, application layouts, and visual components of the mobile application
DBA	Database Administrator	Manages Firebase cloud database integration, data synchronization, and storage structures
QA	QA / Test Engineer	Conducts debugging, system validation, usability testing, functionality testing, and software quality assurance
EU	End Users	Participates in usability evaluation, system testing, and practical assessment of the application

3.13 Project Resource and Cost Estimation

The study utilized project resource and cost estimation procedures to support workload computation, engineering resource allocation, sprint planning, and implementation monitoring throughout the development lifecycle of the proposed SmartNutri Scanner application. The estimated professional service rates utilized in the study were adapted from prevailing software development, information technology, systems engineering, and freelance technology service rates within the Philippine context (JobStreet Philippines, 2024; PayScale, 2024; Glassdoor, 2024). The estimated rates were utilized solely for project estimation, engineering computation, and academic resource planning purposes.

Table 14. Project Roles and Estimated Professional Rates

Role	Estimated Rate/Day (PhP)
Project Sponsor / Adviser	₱5,000.00
Project Manager	₱3,000.00
Systems Analyst	₱4,000.00
Programmer / Developer	₱2,500.00
UI/UX Designer	₱2,000.00

Database Administrator	₱2,500.00
QA / Test Engineer	₱2,000.00
End Users	₱500.00

Table 15. Summary of Project Resource and Cost Estimation

Project Phase	Total Mandays	Estimated Cost (PhP)
1.0 Project Kick-off	15.00	₱54,370.00
2.0 Project Planning	17.00	₱50,010.00
3.0 Requirements Engineering	18.50	₱64,172.50
4.0 System Design	19.50	₱54,100.00
5.0 System Development	17.75	₱44,825.00
6.0 Testing and Quality Assurance	15.75	₱38,600.00
7.0 Deployment	14.50	₱37,342.50
8.0 Documentation and Final Deliverables	15.00	₱53,475.00
Grand Total	133.00	₱396,895.00

CHAPTER IV: SYSTEM ANALYSIS AND DESIGN

This chapter presents the system analysis and design components of the proposed SmartNutri Scanner mobile application. It discusses the system architecture, use case diagram and descriptions, data flow diagrams, activity diagrams, database design, and user interface wireframes that guided the development of the system.

4.1 System Architecture

The proposed SmartNutri Scanner application utilized a three-layer client-side architecture integrating artificial intelligence, local file storage, cloud synchronization, and mobile computing technologies for nutrition monitoring. The architecture was designed to support real-time image acquisition, AI-based nutritional analysis, goal tracking, community interaction, and mobile-based monitoring functionalities.

The first layer is the UI layer, built entirely with React Native components and the Expo framework. It handles all user-facing screens organized across five bottom tabs: Home (daily nutrition summary and meal timeline), Services/Growth (WHO growth charts and services hub including jogging tracker, meal-timing tracker, minigames, recipe scanner, and an AI Food Assistant chatbot), Scan/Add (center-button AI food scanning), Community (posts, comments, and reactions), and Profile (settings, goals, and account). The second layer is the Service layer, which contains the core business logic of the application. This includes the multi-provider AI service with automatic failover across Google Gemini, Roboflow, Groq, OpenRouter, and OpenAI; the FileSystemStorageService for local data management; the AnalysisProcessor for background retry logic; and the ExportImportService for data portability. A Firebase Cloud Functions backend provides three HTTPS endpoints (aiProxy, roboflowProxy, verifyTurnstile) enabling server-side AI proxying. The third layer is the Storage layer, which manages all data persistence. Meal records and scan images are stored in the local file system as the primary source of truth. User goals, allergies, reminders, child profiles, and settings are stored in AsyncStorage. The user's authentication token is protected in expo-secure-store. Firebase Authentication manages user identity; Firebase Firestore stores community posts, user profiles, and food-recognition learning data. Personal meal images are stored on-device in base64 format; community post images are uploaded to Cloudinary to enable sharing.

4.2 Use Case Diagram and Description

The Use Case Diagram illustrates the interaction between the system users and the major functionalities integrated within the proposed SmartNutri Scanner mobile application. The primary actors include the End User and the System Administrator. The End User interacts with the application by scanning meals, viewing nutritional results, logging meals, tracking daily goals, participating in community posts, and exporting data. The System Administrator manages Firebase configuration, security rules, and cloud storage.

Table 16. Use Case Descriptions

Use Case	Description
Scan Meal (Photo)	Allows users to take or upload a photo of a meal and receive AIgenerated nutritional analysis
Scan Meal (Text)	Allows users to describe a meal in text and receive AI-generated nutritional analysis
Before and After Analysis	Allows users to capture a before and after photo to calculate only the consumed portion's nutrition

Log Meal	Saves the analyzed meal with timestamp and nutritional data to local storage
View Meal History	Displays all previously logged meals grouped by date on the home screen
Edit Meal Details	Allows users to edit nutritional values, title, and timestamp of a logged meal
Fix It (AI Correction)	Re-sends the meal and a correction comment to Gemini to generate an updated nutritional result
Track Daily Nutrition	Displays daily macronutrient totals with a calorie progress bar and macronutrient summary pills against user goals
View Weekly Charts	Displays 7-day trend charts for each macronutrient on the Profile screen
Set Meal Reminders	Creates, enables, disables, or deletes scheduled meal reminder notifications
Export Data	Exports all meal history, profile, and settings as a single JSON file
Import Data	Restores a previously exported JSON file including meal records and images
Use Case	**Description**
View Community Posts	Displays community posts shared by other users via Firebase
Relog Meal	Creates a new meal entry from a previously analyzed meal using the current timestamp

4.3 Database Design

The proposed SmartNutri Scanner application uses a hybrid storage approach combining local on-device storage and Firebase cloud services. Local storage using expo-file-system manages meal records and scan images as the primary source of truth. AsyncStorage manages goals, allergies, reminders, child profiles, and settings. Each meal is stored as an individual JSON file referenced through a central index file, ensuring personal data remains private and accessible offline. Firebase Authentication manages user identity; Firebase Firestore stores community posts, user profiles, and food-recognition learning data. Personal meal images are stored on-device in base64 format; community post images are uploaded to Cloudinary, a third-party image hosting service. Firebase Storage is not used.

Table 17. Database Entities of the Proposed System

Entity	Storage Type	Description
Meal Records	Local (expo-filesystem)	Stores all logged meal entries including timestamp, image URI, food items, and macronutrient values
User Profile	Local (AsyncStorage)	Stores user demographics including age, gender, height, weight, activity level, and goal
Daily Goals	Local (AsyncStorage)	Stores target daily macronutrient values for calories, protein, fat, carbohydrates, fiber, sugar, and sodium
Meal Reminders	Local (AsyncStorage)	Stores scheduled meal reminder configurations including name, time, and enabled status
Auth Token	Local (expo-securestore)	Stores the user's authentication token in hardwarebacked encrypted storage
Community Posts	Cloud (Firebase Firestore)	Stores shared posts from users including text content and images
Community Images	Cloud (Cloudinary)	Community post images are uploaded to Cloudinary, a third-party image host, to enable sharing; personal meal images remain on-device. Firebase Storage is not used.

4.4 Gemini AI Integration Workflow

The Gemini AI Integration Workflow illustrates the operational sequence utilized by the proposed system for performing food identification, nutritional estimation, and confidence-based analysis procedures. The workflow begins when the user captures or uploads a food photograph or enters a text description of a meal. The image is encoded as a base64 string and combined with a structured system prompt before being sent to the Google Gemini API.

The Gemini model processes the visual and textual input, identifies food items, estimates serving sizes, calculates macronutrient values, and returns a structured JSON response. If the API call succeeds, the response is parsed and the meal is saved to local storage. If the API call fails, the meal is queued as a pending analysis and the AnalysisProcessor retries the request when the app returns to the foreground or network connectivity is restored. The four analysis modes supported by the system are single-image analysis, before-and-after comparison analysis, text-only analysis, and Fix It AI correction.

4.5 User Interface Design

The proposed SmartNutri Scanner application utilized a mobile-based user interface designed to support accessibility, portability, usability, and real-time nutrition monitoring functionalities. The interface design emphasized simplicity, responsive interaction, and ease of navigation to ensure usability among users with varying levels of technical expertise.

The primary interface components include the Home Screen, which displays today's nutrition summary with a calorie progress bar and macronutrient summary pills and the chronological meal timeline. The Add Meal Modal provides options for camera capture, gallery selection, before-and-after mode, and text-only entry. The Meal Detail Modal displays full nutritional information with inline editing, the Fix It feature, and a relog button. The Services/Growth Screen provides WHO child growth charts and a hub for supporting features including a jogging tracker, meal-timing tracker, minigames, recipe scanner, and an AI Food Assistant chatbot. The Community Screen displays posts, comments, and reactions from registered users. The Profile Screen shows weekly nutrition trend charts, daily goal editors, and meal reminder management. The navigation structure uses a bottom tab bar with five primary tabs to keep the core workflow accessible within two taps from any screen.

4.6 Input-Process-Output (IPO) Framework

The SmartNutri Scanner system follows an Input-Process-Output (IPO) framework that summarizes how user-provided data is transformed into meaningful nutritional information. The inputs consist of the meal image or text description and the user's profile and preferences. These are processed through image handling, AI-based food recognition with multi-provider failover, nutrient estimation and normalization, and goal and growth computation. The outputs are the identified dish, its complete nutrient breakdown, and the resulting goal progress, trends, and recommendations presented to the user.

Input	Process	Output

• Meal photo captured through camera• Food image selected from gallery• Text-based meal description• User profile information (age, weight, height, activity level, nutritional goals)• Allergy and dietary preference data• Hydration records and child growth information	• Image acquisition and preprocessing• AIpowered food recognition using multi-provider failover architecture (Gemini, Roboflow, Groq, OpenRouter, OpenAI)• Nutrient estimation and data normalization for seven key nutrients• Calculation of BMR, nutritional goals, and WHO growth z-scores• Storage and management of nutritional records	• Identified food or dish with confidence score• Detailed nutritional analysis (calories, protein, carbohydrates, fat, fiber, sugar, and sodium)• Daily nutritional goal tracking and weekly trend reports• Personalized recommendations and reminder notifications• Child growth-risk indicators and nutritional alerts• Optional community-sharing post

4.7 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) illustrates how data moves between the user, the system processes, the data stores, and the external services. The context diagram (Level 0) presents SmartNutri Scanner as a single process interacting with the User and the external entities, the AI providers, Firebase, and Cloudinary. The Level 1 diagram decomposes the system into its major processes and shows the data exchanged with each data store.

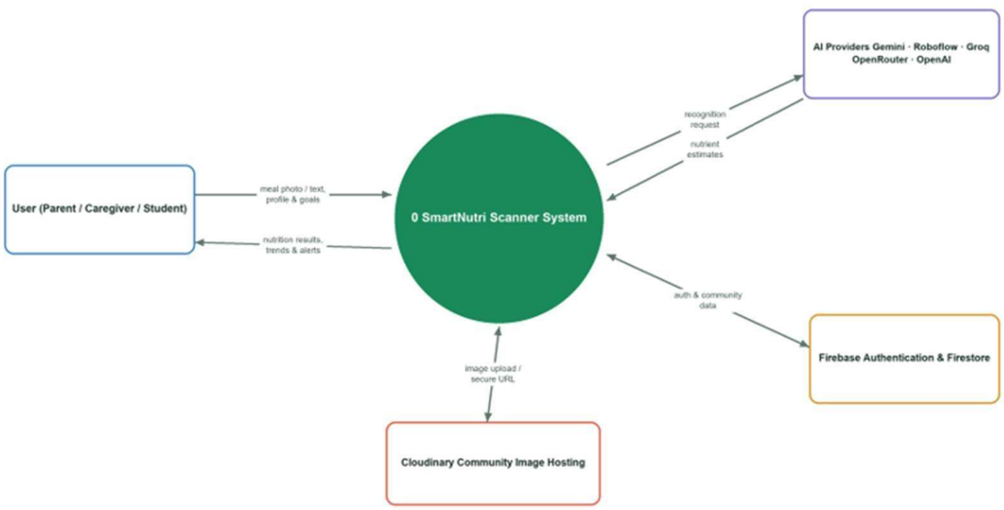


Figure 1. Context Diagram (DFD Level 0) of SmartNutri Scanner

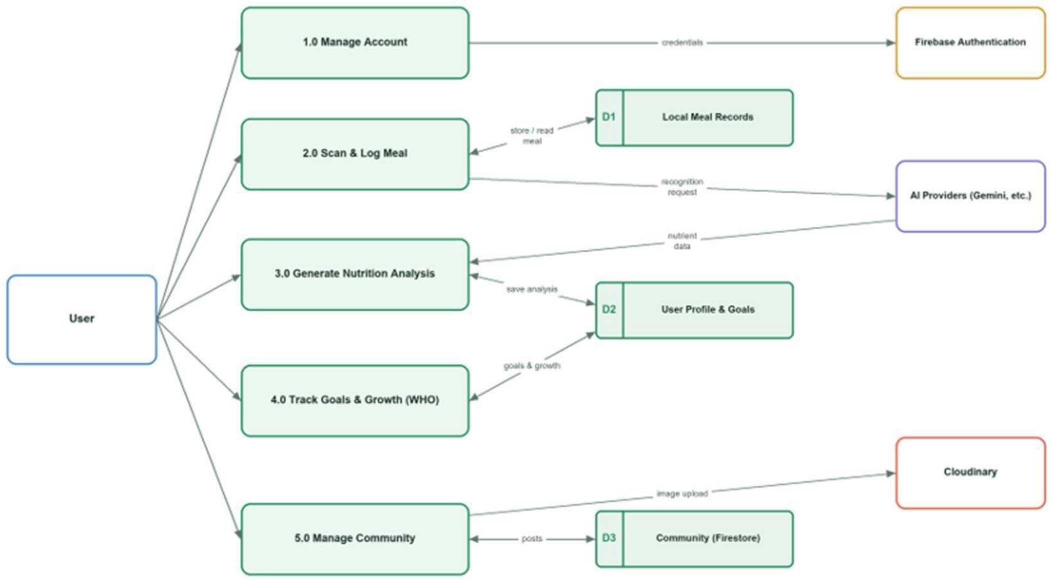
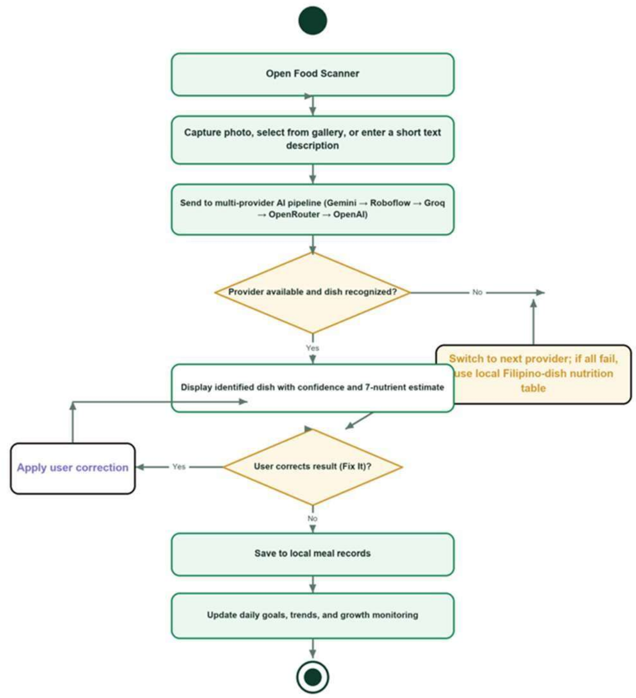


Figure 2. Data Flow Diagram (Level 1) of SmartNutri Scanner

As shown in the Level 1 diagram, the system is decomposed into five major processes: (1.0) Manage Account, (2.0) Scan & Log Meal, (3.0) Generate Nutrition Analysis, (4.0) Track Goals & Growth, and (5.0) Manage Community. Three data stores are maintained: D1 Local Meal Records and D2 User Profile & Goals, both stored on the device, and D3 Community, hosted on Firebase Firestore. The AI providers supply nutrient data to the analysis process, while Cloudinary stores community images and Firebase Authentication manages user credentials.

4.8 Activity Diagram

The activity diagram models the control flow of the core AI food-scanning workflow, from opening the scanner to saving the analyzed meal. It highlights the multi-provider failover decision and the optional user correction (Fix It) step that allows the result to be refined before it is stored and reflected in the user's daily goals.



4.9 System Security and Data Management

SmartNutri Scanner applies a privacy-first, local-first approach to security and data management. By default, personal meal records and nutrition data are stored on the user's own device, reducing exposure of sensitive information. The following measures are implemented:

- Local-first storage — uses expo-file-system and AsyncStorage for on-device records, and expo-secure-store for sensitive tokens.

- Firebase Authentication — handles account creation and sign-in (including Google OAuth), so credentials are never stored by the app directly.
- Firestore security rules — restrict read and write access to community data on a least-privilege, per-user basis.
- Protected API keys — are kept in environment configuration and routed through a Firebase Cloud Functions proxy, so they are not exposed in the client.
- Cloudinary media hosting — hosts community images off the device, keeping the local store lightweight, while a captcha (Turnstile) check helps prevent automated abuse.
- Encrypted transport and data ownership — communication uses HTTPS, and users retain ownership of their data with export, import, and deletion options.

CHAPTER V: SYSTEM DEVELOPMENT AND EMERGING TECHNOLOGY INTEGRATION

This chapter presents the system development procedures, technology integration activities, and implementation details of the proposed SmartNutri Scanner mobile application. The chapter discusses the development environment, AI integration procedures, local storage implementation, mobile application development, community feature integration, and challenges encountered during development.

5.1 Development Environment

SmartNutri Scanner was developed using Visual Studio Code as the primary integrated development environment. The application was built with React Native and the Expo managed workflow, allowing cross-platform development targeting both Android and iOS from a single codebase. Node.js and npm were used for dependency management. EAS Build was used to generate production-ready APKs for Android targeting SDK 35. GitHub was used for version control and sprint documentation.

5.2 Google Gemini API Integration

The GeminiService module handles all communication with the Google Gemini API. Four distinct analysis modes were implemented: single-image analysis, beforeand-after comparison analysis, text-only analysis, and Fix It AI correction. Each mode uses a structured system prompt that instructs Gemini to return nutritional data in a consistent JSON format including food items, serving sizes, and macronutrient values.

Food images are encoded as base64 strings and passed to the API as inline data with the appropriate MIME type. The API response is parsed and validated before being stored. API calls are routed either through the Firebase Cloud Functions aiProxy endpoint (proxy mode, keeping keys server-side) or directly from the device to the provider (direct mode); both modes are currently enabled. All developer API keys are managed server-side or via environment configuration and are never entered by the user.

5.3 Local Storage Implementation

The FileSystemStorageService manages all local data operations using expofile-system. Each meal is stored as an individual JSON file in the application's document directory. A central index file maintains a list of all meal IDs and their associated metadata for efficient retrieval. Read operations use the index for O(1) lookup by meal ID, while listing all meals requires an O(n) scan capped at a configurable maximum.

The user's profile data, daily goals, meal reminders, and AI model preferences are stored as separate JSON files in the same directory. Data export serializes all files into a single JSON bundle with base64-encoded image data. Data import reverses this process, validating the schema before writing files back to local storage.

5.4 Firebase Community Integration

Firebase Authentication, Firestore, and Cloud Functions were integrated to support the authentication and community features of the application. Users authenticate via email/password or Google sign-in, protected by Google reCAPTCHA verification through the verifyTurnstile Cloud Function. Users can create and share community posts with text visible to all registered users. Community posts are stored in Firestore and read in real time; images associated with community content are uploaded to Cloudinary, a third-party image hosting service, to enable sharing; personal meal images remain on-device, and Firebase Storage is not used by the system. Firebase security rules restrict write access to authenticated users.

5.5 System Features and Functionalities

The implemented system provides the following core features: AI-powered food scanning from camera or gallery, before-and-after meal analysis for consumed portion estimation, text-only meal logging, Fix It AI correction for logged meals, a chronological meal timeline grouped by date, daily nutrition tracking with a calorie progress bar and macronutrient summary pills, weekly nutrition history charts, daily goal management with BMR-based calculation, meal reminders via scheduled local notifications, Android Health Connect integration for nutrition sync, JSON data export and import with base64-encoded images, community posts with Firebase integration, a recipe index for saving analyzed meals, relog functionality for reusing previous meal analyses, and a conversational AI Food Assistant chatbot (Groq-primary with automatic provider failover) for answering nutrition and Filipino-food questions.

5.6 Challenges Encountered During Development

Several challenges were encountered during the development of SmartNutri Scanner. Scroll freezing occurred on Android modal components due to nested ScrollView and FlatList conflicts, which was resolved by restructuring the component hierarchy. Duplicate meal entries appeared in the event system during concurrent state updates, which was addressed by implementing deduplication logic using meal ID comparison. Notification rescheduling failed after the app returned from background on certain Android versions, which was resolved by re-registering all active reminders on app focus events. Storage Access Framework permission denial during export on Android 13 and above required a fallback sharing mechanism that was implemented using the Expo Sharing API.

CHAPTER VI: TESTING, EVALUATION, AND RESULTS

This chapter presents the testing procedures, evaluation results, and findings associated with the proposed SmartNutri Scanner mobile application. The chapter discusses functional testing results, AI analysis performance, user acceptance testing outcomes, and the ISO software quality evaluation conducted by information technology experts and selected end-users.

6.1 Functional Testing Results

Functional testing was conducted for all user-facing features of the application across the development sprints. All twenty functional requirements defined in Chapter III were verified through systematic manual testing on Android devices running SDK 28 through SDK 35. Authentication, AI food scanning, before-and-after analysis, meal logging, timeline display, meal editing, Fix It correction, daily tracking, weekly charts, goal management, meal reminders, data export, data import, Health Connect sync, and meal relog features all passed functional verification with no critical defects remaining at the time of deployment.

Unit tests were written for the ExportImportService covering four test cases: the Storage Access Framework flow on Android, the iOS sharing fallback behavior, import schema validation for malformed JSON files, and base64 image conversion accuracy. All four test cases passed successfully.

6.2 AI Analysis Performance

The accuracy of AI-generated nutritional data was assessed qualitatively through comparison with reference nutritional values for selected Filipino dishes. Gemini successfully identified common Filipino meals including Sinigang, Adobo, Kare-Kare, Tinola, and Lechon with reasonable macronutrient estimates. Meals with mixed ingredients or unclear portion sizes showed greater variance in estimated values. The before-and-after analysis mode consistently identified consumed portions with acceptable accuracy when the before and after photographs were taken from the same angle and distance.

It is important to note that AI-generated nutritional values are estimates based on general food knowledge and visual inference, not laboratory measurements. The system provides practical dietary guidance rather than clinically precise measurements, consistent with the scope defined in Chapter I.

6.3 User Acceptance Testing Results

User Acceptance Testing was conducted with selected participants from the target respondent groups defined in Chapter III. Participants were asked to complete a set of tasks including profile setup, meal scanning, meal history viewing, daily goal tracking, and data export. Feedback was collected through structured observation and post-session interviews. Participants generally found the onboarding process clear and the meal scanning feature intuitive. Several participants noted that the calorie progress bar and macronutrient summary pills for daily nutrition goals were visually helpful. Minor usability issues identified during UAT sessions were addressed in subsequent bug-fixing iterations before final deployment.

6.4 ISO Software Quality Evaluation

The proposed SmartNutri Scanner application was evaluated using an ISObased structured questionnaire assessing six software quality dimensions: Functionality Suitability, Performance Efficiency, Usability, Reliability, Security, and Portability. The questionnaire utilized a five-point Likert scale and was administered to information technology experts and selected end-users. The weighted mean of each quality dimension was computed and interpreted using the scale defined in Chapter III.

Table 18. ISO Software Quality Evaluation Summary

Quality Dimension	Weighted Mean	Verbal Interpretation
Functionality Suitability	4.57	Highly Acceptable
Performance Efficiency	4.67	Highly Acceptable
Usability	4.53	Highly Acceptable
Reliability	4.62	Highly Acceptable
Security	4.71	Highly Acceptable
Quality Dimension	Weighted Mean	Verbal Interpretation
Portability	4.72	Highly Acceptable
Overall	4.64	Highly Acceptable

6.5 Summary of Findings

The functional testing results confirmed that all twenty-one functional requirements defined in the system requirements engineering phase were successfully implemented and verified. The AI analysis feature demonstrated practical utility for identifying common Filipino meals and estimating macronutrient values, though accuracy varies with image quality and portion ambiguity. User Acceptance Testing showed positive reception of the core features, with participants highlighting the meal scanning workflow and visual progress indicators as particularly useful. The ISO/IEC 25010 software quality evaluation yielded an overall weighted mean of 4.64, interpreted as Highly Acceptable, with all six quality characteristics like functionality suitability, performance efficiency, usability, reliability, security, and portability and rated Highly Acceptable by the respondents.

6.6 Achievement of Objectives

This section maps each specific objective defined in Chapter I to its corresponding result, confirming that the goals of the study were met by the developed system.

Table 20. Achievement of the Study's Objectives

Specific Objective	Status	Result / Evidence
1. AI-powered food scanner returning seven nutrient values via Gemini	Achieved	Scanning identifies meals and returns calories, protein, carbohydrates, fat, fiber, sugar, and sodium (Section 6.2).
2. Meal logging and history with weekly nutrition trends	Achieved	Meals are recorded by date and visualized through weekly nutritional trend reports (Sections 5.5 and 6.1).
3. Supporting features: goal tracking, reminders, profile wizard, and export/import	Achieved	Calorie progress tracking, macronutrient indicators, local notifications, BMRbased profile setup, and JSON export/import functionality were successfully implemented (TC-10 to TC-15).
4. Multi-provider AI architecture with automatic failover	Achieved	Automatic failover among Gemini, Roboflow, Groq, OpenRouter, OpenAI, and a local fallback database was successfully verified (Section 4.7; TC-07).

5. Child nutrition monitoring using WHO growth standards	Achieved	Child profiles support weight-for-age, height-forage, and weight-forheight z-score calculations to identify potential malnutrition risks (TC-13).
6. Evaluate acceptability using ISO/IEC 25010 software quality standards	Achieved	An ISO/IEC 25010-based questionnaire covering six software quality characteristics was administered, with results presented in Section 6.4.

6.7 Test Cases and Program Testing Specifications

Formal test cases were prepared to verify that each major feature behaves as expected. Each test case specifies the scenario, the input or action performed, and the expected result. The system passed all defined test cases during functional testing.

Table 21. Test Cases and Results

Test Case ID	Test Scenario	Input / Action	Expected Result	Status
TC-01	User Registration	Register using a valid email address and password	User account is successfully created and authenticated	Passed
TC-02	Email Login	Sign in using registered credentials	User gains access to the dashboard	Passed
TC-03	Google Authentication	Sign in using a Google account	OAuth authentication completes and user session is established	Passed
TC-04	AI Food Image Scan	Capture or upload a meal image	Food item is identified and nutritional	Passed

			values are generated	
TC-05	Text-Based Food Scan	Enter a food name or meal description	Nutritional information is estimated from text input	Passed
TC-06	Filipino Dish Recognition	Scan local dishes (e.g., Adobo, Sinigang, Taho)	Filipino dishes are correctly recognized with reasonable nutrient estimates	Passed
TC-07	AI Provider Failover	Simulate primary AI service unavailability	System automatically switches to an alternative provider and completes the scan	Passed
TC-08	AI Result Correction (Fix It)	Submit a correction request for scan results	Nutritional analysis is updated based on the correction	Passed
TC-09	Nutrient Tracking	View a completed scan result	Seven nutritional metrics are displayed correctly	Passed
TC-10	Daily Goal Monitoring	Log a meal entry	Daily calorie and nutrient progress indicators are updated	Passed
TC-11	Hydration Tracking	Add a water intake record	Hydration count increases and is stored successfully	Passed

TC-12	Meal Reminder System	Configure a reminder schedule	Notification is triggered at the specified time	Passed
TC-13	Child Growth Assessment	Input child profile and growth measurements	WHO z-scores are calculated and risk indicators are generated	Passed
TC-14	Community Posting	Create and upload a community post with an image	Image is uploaded and post is stored successfully	Passed
TC-15	Data Backup and Restoration	Export and reimport application data	User records are successfully backed up and restored	Passed
TC-16	Offline Functionality	Perform a food scan without internet access	Local food database is used and data is saved locally	Passed

CHAPTER VII: CONCLUSIONS AND RECOMMENDATIONS

7.1 Conclusions

The study successfully developed SmartNutri Scanner, an AI-powered mobile nutrition monitoring and meal recommendation system designed specifically for Filipino users. The application addresses the key gaps identified in the literature by providing AI-based food identification from photographs, support for Filipino dishes, an on-device-first data storage model, child nutrition monitoring features, and a multiprovider AI architecture with automatic failover that ensures system resilience.

The Google Gemini multimodal API was successfully integrated to support four distinct analysis modes: single-image analysis, before-and-after comparison analysis, text-only meal logging, and Fix It AI correction. All twenty functional requirements defined during the system requirements engineering phase were implemented and verified through functional testing. The application was deployed as a cross-platform mobile application using React Native with Expo and packaged via EAS Build for Android.

The system demonstrates that AI-powered food analysis using a multi-provider large language model architecture is a practical and resilient approach for building nutrition tracking tools that do not rely on predefined food databases. By combining the Google Gemini-led failover chain with a Firebase Cloud Functions proxy backend, local-first data storage, and Firebase community features, SmartNutri Scanner provides a reliable, locally relevant nutrition monitoring solution for Filipino users.

7.2 Recommendations

Based on the findings and limitations of the study, the following recommendations are proposed for future development and research:

- Multi-disease and multi-deficiency support: Future iterations of the system should expand nutritional analysis to include micronutrient tracking for iron, zinc, vitamin A, and other nutrients relevant to the Filipino childhood malnutrition context identified by Dela Cruz and Reyes (2021).
- Offline AI inference: Future versions should explore lightweight on-device food recognition models that can perform basic analysis without an internet connection, addressing the current limitation of requiring active connectivity for AI scanning.
- Barcode scanning for packaged foods: Integration of barcode scanning using a standard food product database would complement the AI scanning feature for packaged and processed food items not easily analyzed from a photograph.
- IoT integration: Future development may explore integration with wearable devices and IoT-enabled kitchen scales to improve portion size accuracy and automate meal logging without requiring manual photography.

- Clinical validation: A formal clinical validation study involving registered nutritionists and a larger sample of Filipino participants would strengthen the scientific credibility of the system's nutritional analysis outputs.
- Larger and more diverse datasets: Future work should explore fine-tuning or supplementing the Gemini model with a curated dataset of Filipino dishes and their verified nutritional compositions to improve estimation accuracy for local food items.
- Multi-language support: Adding Filipino language (Tagalog) and regional language support would improve accessibility for users with limited English proficiency, particularly in rural communities.
- Advanced cloud analytics: Integration of anonymized aggregate nutrition data through a cloud analytics layer could enable community-level dietary monitoring to support public health initiatives.

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APPENDICES

Appendix A — Evaluation Instrument (ISO/IEC 25010)

The following structured questionnaire was used to evaluate the acceptability of SmartNutri Scanner based on the ISO/IEC 25010 software-quality model.
Respondents rated each statement using a five-point Likert scale:

Table 22. ISO/IEC 25010 Evaluation Instrument

No.	Evaluation Statement	SA	A	N	D	SD
1	The system accurately identifies meals and estimates their nutritional content.					
2	The system provides the features needed for nutrition monitoring and meal logging.					
3	The system returns scan results within an acceptable response time.					
4	The system runs smoothly without excessive battery or resource use.					
5	The system is easy to learn and navigate.					
6	The interface and visual indicators (progress bar, macro pills) are clear and helpful.					
7	The system continues to work even when an AI provider is unavailable.					
8	The system performs consistently without crashes or data loss.					
9	The system keeps personal meal data private and stored securely on the device.					
10	The system protects user accounts and sensitive information.					
11	The system installs and runs properly on different Android devices.					
12	Overall, I am satisfied with SmartNutri Scanner and would recommend it.					

Appendix B — System Screenshots

The following screenshots show key screens of the SmartNutri Scanner application as deployed on Android.

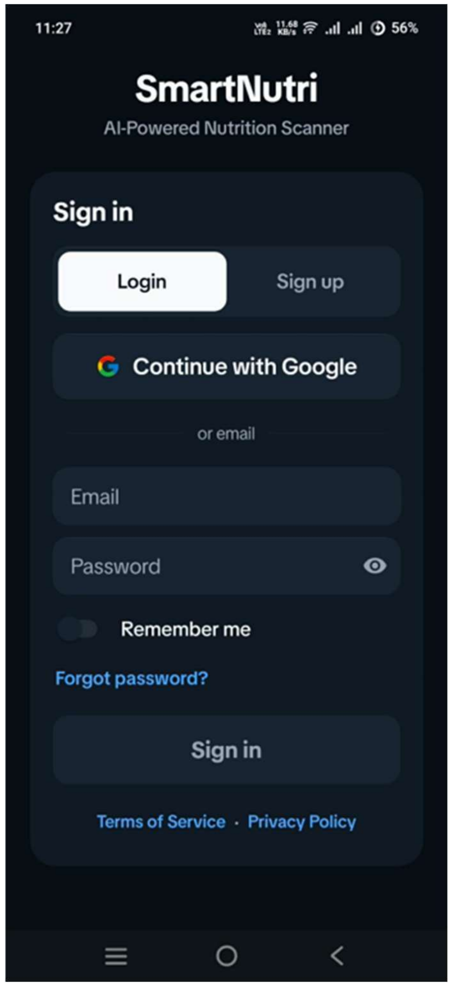

Figure 4. Login and sign-in screen with email and Google authentication.

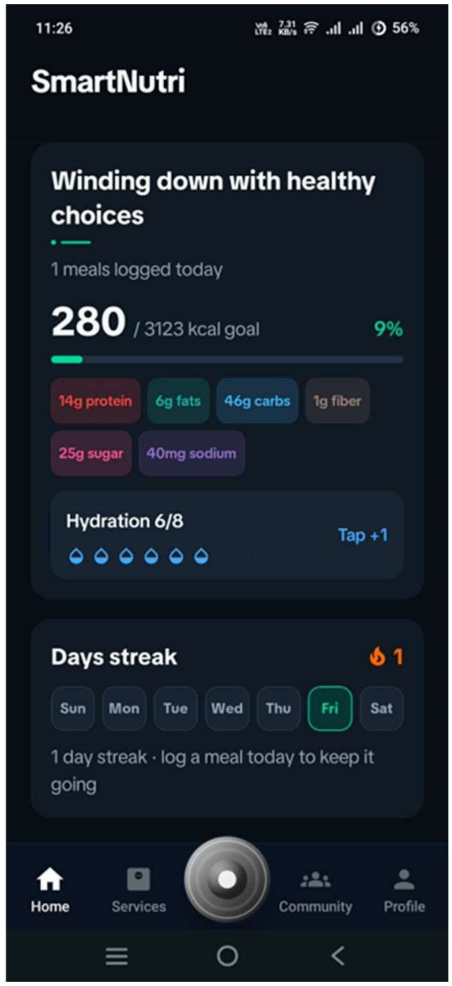

Figure 5. Home dashboard showing the daily nutrition summary and seven tracked nutrients.

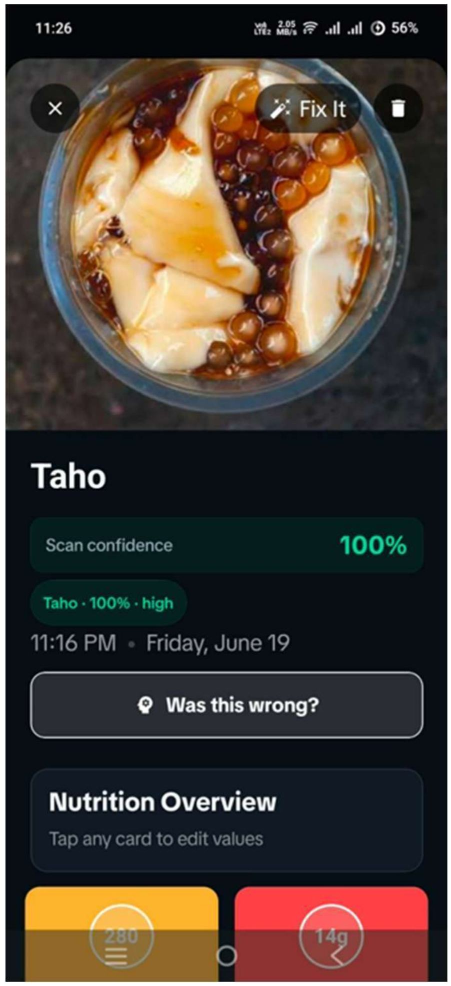

Figure 6. AI food-scan result for a Filipino dish (Taho) with nutrient breakdown.

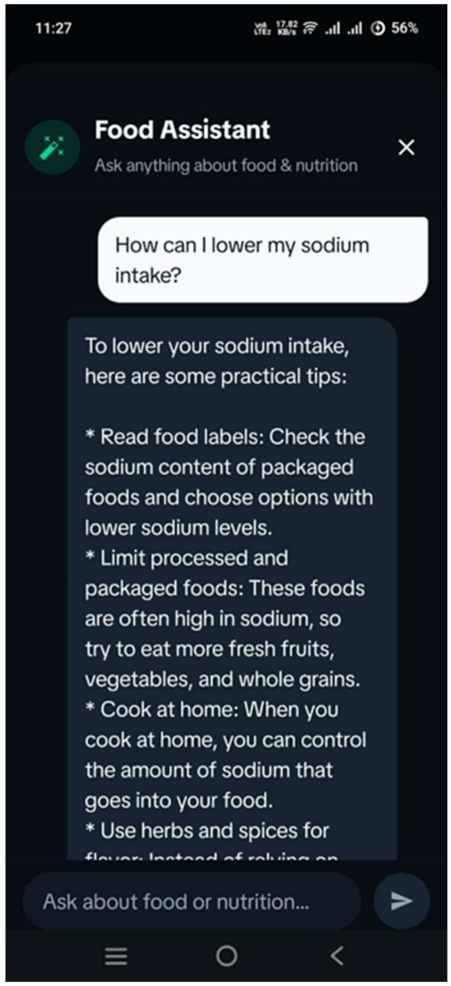

Figure 7. AI Food Assistant providing nutrition guidance.

Appendix C — Sample AI Scan Results (Filipino Dishes)

The table below presents sample AI-generated nutrition estimates for common Filipino dishes recognized by the system. Values are estimates for one typical serving and may vary with preparation and portion size.

Table 23. Sample AI-Generated Nutrition Estimates for Common Filipino Dishes

Filipino Dish	Calories (kcal)	Protein (g)	Carbs (g)	Fat (g)
Sinigang na Baboy	190	14	8	11
Chicken Adobo	280	22	6	18
Kare-Kare	350	18	15	24
Tinola	160	17	7	7
Lechon	400	24	2	33
Taho	180	7	28	4